Remote sensing image semantic segmentation network with feature comparison and parallel fusion

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***Abstract*—We presented a novel network architecture integrating Swin Transformer and Convolutional Neural Networks (CNNs) for enhanced local and global feature extraction and fusion. Combining Swin Transformer and CNNs within a dual encoder setup and featuring a novel Feature Fusion Module (FFM) and Contrast Attention Module (CAM), the network adeptly integrates local and global information, improving feature discernibility and segmentation accuracy. By incorporating a joint loss function of soft cross-entropy and Dice loss, the model shows notable results on the Beijing Land Use dataset, exceeding several top methods in semantic segmentation efficacy.**

Keywords-Semantic Segmentation, Remote Sensing Images, Feature Fusion, Contrast Attention

I. INTRODUCTION

With the rapid advancement of space and sensor technologies, acquiring high-quality remote sensing images has become easier, which has propelled the development of remote sensing image semantic segmentation technology. This technology has wide-ranging applications in urban planning, disaster response, and agriculture. However, small objects in remote sensing images, along with their high similarity and overlapping, create challenges for accurate semantic segmentation. In recent years, deep learning methods, particularly convolutional neural networks and transformers, as well as their hybrid models, have shown substantial progress by effectively extracting local features and expanding the receptive field. Despite these improvements, there are still deficiencies in understanding the global context and in local accuracy. To overcome these limitations, our research has explored the application of feature fusion and attention mechanisms to improve the semantic segmentation of remote sensing images. Our experiments indicate that in-depth research into how to integrate local and global information effectively is crucial for enhancing the accuracy and robustness of remote sensing image semantic segmentation.

II. RELATED WORK

A. CNN and variants

CNN variant models typically utilize convolution, pooling, and encoder-decoder structures to build the network, while fully convolutional networks are designed to process images of different sizes in a single forward pass [1]. Examples include models such as ResNet [2], UNet [3], and A2FPN [6]. These models utilize upsampling and downsampling processes to modify the resolution of feature maps, allowing features to be captured at various scales. However, semantic categories in remotely sensed images may present similar size, texture, and spectral properties, thus complicating their differentiation. For example, urban areas, forested areas, and bodies of water may have similar textures and colors, leading to potential misclassification. Additionally, obstacles such as clouds or man-made structures may introduce uncertainty into semantic interpretation. Therefore, the broader global context and fine-grained spatial features are crucial for semantic reasoning, enabling models to navigate the inherent complexity and variability of remote sensing images.

B. Vision Transformer

Transformers, originally designed for language processing, have been repurposed for computer vision by treating images as token sequences [4]. This enables global self-attention or cross-attention, aiding image segmentation. Researchers have proposed combining transformers with CNNs to provide localization cues. For example, integrating ViT with UNet [5] leverages the strengths of both models, resulting in more accurate medical image segmentation. Another innovative approach is the Swin Transformer [10]. By introducing a sliding window strategy, it addresses the globalizing tendency of self-attention mechanisms in 2D image tasks, offering better local information processing capabilities and effectively reducing computational complexity.

III. METHOD

A. Network Architecture

Figure 1 presents the overall network architecture, the encoder is divided into Swin-T encoding branch and Resnet encoding branch. The output of each stage of Swin-T goes through a CAM, and then is merged with the ResNet of each stage through FFM. The decoder is designed with an upsampling structure similar to unet.

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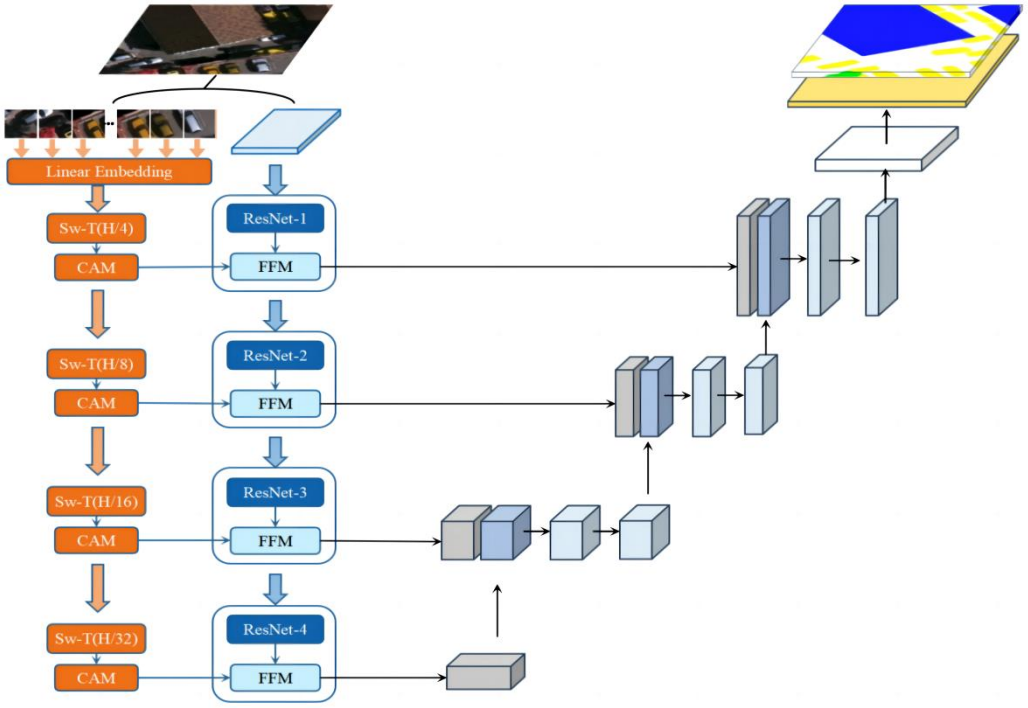


Figure 1 Framework.

B. Feature Fusion Module

The Feature Fusion Module FFM designed in this article aims to use the Swin-T branch encoder to enhance the network's ability to identify local information with spatial constraints, and can simultaneously capture the complex relationship dependencies that exist between different channels. Although features may show similar distribution patterns in space, their representation on different channels may vary significantly. Previous studies have identified the effects of channel encoding in enhancing feature perception. FFM focuses on amplifying the intensity of more critical channels over the entire feature map by building channel that converge with local features obtained from the ResNet encoder.

By using deformable convolutions in ResNet encoding, FFM enhances the flexibility of the network to adapt to different shapes of objects in remote sensing images. It refines these local features through the Swin-T encoder and further unifies more global discriminative features, potentially improving the segmentation accuracy of ground objects that exhibit high similarity in RS images.

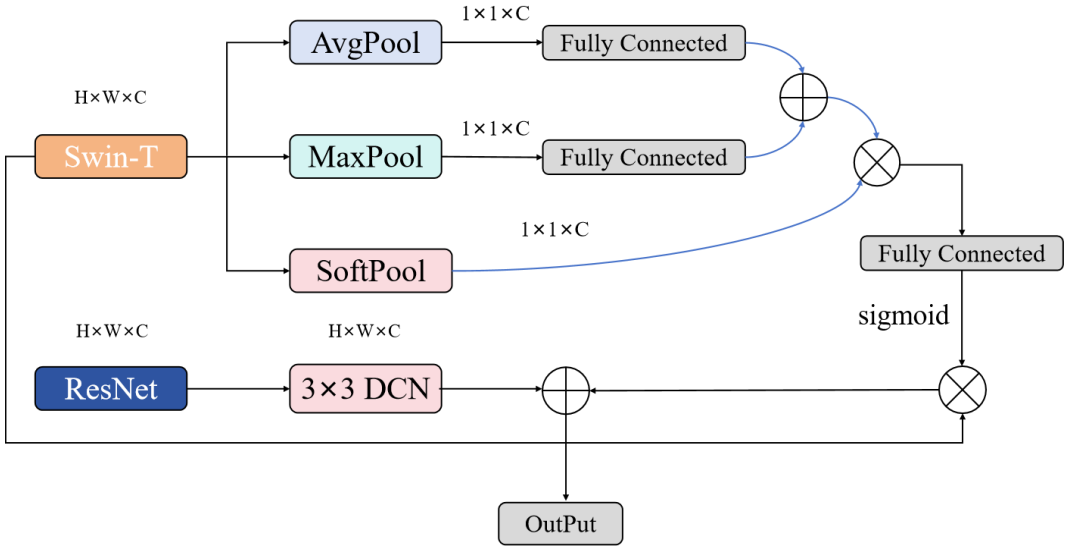


Figure 2 Feature fusion module

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Figure 2 illustrates the FFM, which employs three pooling strategies in parallel: average pooling, maximum pooling, and soft pooling. The process can be expressed by the following equation (1~3):

$$S_{A\&M} = fc(AvgPool(S_n)) + fc(MaxPool(S_n)) \quad (1)$$

$$S_S = fc(SoftPool(Sn)) \quad (2)$$

$$S_n^{'} = fc(S_{AM} \odot S_S) \quad (3)$$

S_n represents the output of each stages of Swin-T. *fc* means fully connected layer. and $\odot$ indicates element-wise multiplication. $S_n^{'}$ is a feature after channel optimization. By encoding these dependencies, the distinction of features is improved.

C. Contrast Attention Module

Figure 3 illustrates the feature contrast attention module, which integrates multi-scale information and attention mechanisms to enhance feature expression. Dilation convolution can increase the receptive field without losing resolution. Different dilation rates (1, 3, 5, and 7) allowing the network to observe a wider range of contextual information. The subtraction operation of features of different scales helps to highlight the differences between these features, enhancing the contrast of features and improving the segmentation effect in regions with significant feature differences. The merged feature map is further enhanced by a spatial attention mechanism, which improves the network's sensitivity to shadow and occlusion areas in the image.

Contrary to spatial attention, the channel attention mechanism directly processes the merged features. It evaluates the importance of each channel under different receptive fields and adjusts the channel weights accordingly. This module can enhance the model's capability to recognize the boundaries of diverse objects in an image, particularly small objects or intricate details against complex backgrounds.

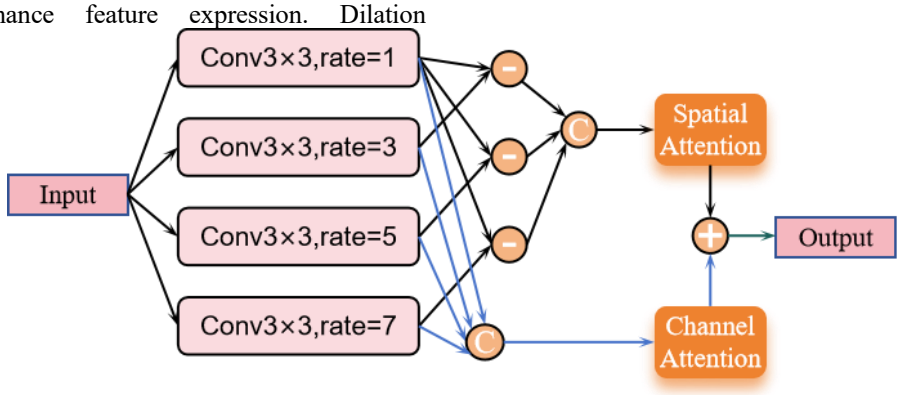


Figure 3 Contrast attention module

D. joint-loss

The soft cross-entropy function works by smoothing the true labels, which helps to compute cross-entropy more effectively. The function's formula is provided in Eq (4), showing how it operates and how smoothing contributes to a more robust solution. This method is applicable to a variety of use cases, demonstrating its versatility.

$$Celoss = -\sum_{i=1}^{C} p_i \log(q_i) \quad (4)$$

$$Diceloss = 1 - \frac{2|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}|} \quad (5)$$

$$softmax(\cdot) = \frac{e^{q_i}}{\sum_{j=1}^{N} e^{q_j}} \quad (6)$$

The probability p_i represents the true label, while q_i is the predicted probability after applying *softmax*(·) output,defined by Eq (6). The Dice loss function offers a remedy for class imbalance, although it can introduce training instability. Defined by Equation (5), the Dice loss employs both *X* (ground truth) and *Y* (predicted). We use a joint loss combining soft cross-entropy and Dice loss to supervise training for the CNN and Swin-T branches, as defined by Eq (7). The balancing coefficient λ between the two is set to 1.

$$L = \lambda_1 \cdot Diceloss + \lambda_2 \cdot Celoss \quad (7)$$

IV. Experiments

A. Dataset

The BLU (Beijing Land-Use) dataset is a high-resolution benchmark dataset designed for multi-class semantic segmentation research on Remote Sensing Images (RSIs). Collected by the Beijing-2 satellite in June 2018, it includes RGB optical images with a Ground Sampling Distance (GSD) of 0.8 meters. The dataset provides fine-grained human annotations for six land-use classes: background/barren, built-up, vegetation, water, agricultural land, and road. These classes are essential for various applications, including environmental monitoring, traffic analysis, and urban and rural management. The BLU dataset stands out for its high spatial resolution, accurate annotations by an experienced team, and diverse scenes spanning urban and rural areas, covering around 150 square kilometers of Beijing. It consists of four large tiles,

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further divided into 64 sub-images, providing a comprehensive training, validation, and testing set for researchers.

B. Experimental setting

Our model was trained on NVIDIA V100-SXM2 GPUs within the PyTorch deep learning framework, using the SGD optimizer. The initial learning rate was set to 0.01, with a momentum of 0.9 and a weight decay of 5*e*-4, following [8]. During training, images and labels were randomly resized with scale factors from 0.5 to 2.0. Further data augmentation techniques such as random scaling, cropping, and flipping were also applied.

C. Evaluation metrics

We adhere to the official guidelines for evaluation metrics, including *F1*, *OA*, and *mIoU*, which serve as key indicators. The formulas are provided by Eqs (8-10).

$$F_1 = 2 \times \frac{P \times r}{P + r} \tag{8}$$

$$IOU = \frac{TP}{TP + FP + FN} \tag{9}$$

$$OA = \frac{TP}{TP + FP + FN + TN} \tag{10}$$

P and *N* represent positive and negative, respectively. *TP* denotes the pixels that are correctly classified as positive, while *TN* denotes those correctly classified as negative. *FP* represents pixels incorrectly predicted as positive, while *FN* represents pixels incorrectly predicted as negative.

D. Results

Quantitative comparison: On the BLU dataset, we conducted a comprehensive evaluation of our method in comparison to LANet [9], A2FPN [6], WiCoNet [7], and DCST [8]. As shown in Table 1, our method demonstrates significant improvements, achieving gains of +2.2%, +1.33%, and +0.41% (mIoU) over LANet, WiCoNet, and DCST, respectively. According to the *F1* and *mIoU*-score evaluation metrics, our method outperforms state-of-the-art models across all categories, except for the vegetable and agricultural classes.

Table 1 Percentage-based comparison results with advanced models on the BLU dataset.

Method	Background	Built-up	Veg	Water	Agriculture	Road	mIoU	F1	OA
LA-Net[9]	73.85	87.30	90.73	86.13	86.8	68.58	70.63	82.23	86.49
A2-FPN[6]	70.28	86.70	89.90	84.00	86.66	64.96	68.22	80.42	85.85
WiCoNet[7]	74.43	88.55	90.94	86.01	87.23	70.21	71.50	82.89	87.35
DCST[8]	75.11	88.51	91.36	86.80	89.49	69.52	72.42	83.46	88.09
ours	75.37	88.92	90.51	87.44	88.12	70.78	72.83	83.52	88.26

Qualitative Comparison: Qualitative Comparison: Figure 4 provides a visual comparison of segmentation results from different models, particularly focusing on LANet and A2FPN. Both models show somewhat blurred segmentation contours, which is quite apparent in the depiction of water bodies and agricultural lands, as marked by the purple box in the figure. This blurring is mainly due to the models' lack of comprehensive global contextual information, which is essential for accurately differentiating landscape features that may look similar in texture or color but differ significantly in spatial arrangement. Furthermore, our network significantly outperforms the WiCoNet and DCST models in terms of building delineation. Our network provides clearer building edges. This improvement is attributed to the integration of the CAM and the FFM within our network architecture. The CAM enhances feature contrasts around object edges and boundaries, crucial in urban settings where buildings often have similar textures and colors, potentially leading to semantic ambiguities. By enhancing contrast at a detailed level, CAM helps in clearly distinguishing individual buildings, even those that are closely spaced.

Moreover, the FFM ensures that while local detail is preserved, the global contextual understanding is enhanced, maintaining the integrity of distinct building outlines. Conversely, models like WiCoNet and DCST, though generally effective, fall short in sharply delineating complex urban structures, resulting in poorer performance in scenarios with intricate building arrangements.

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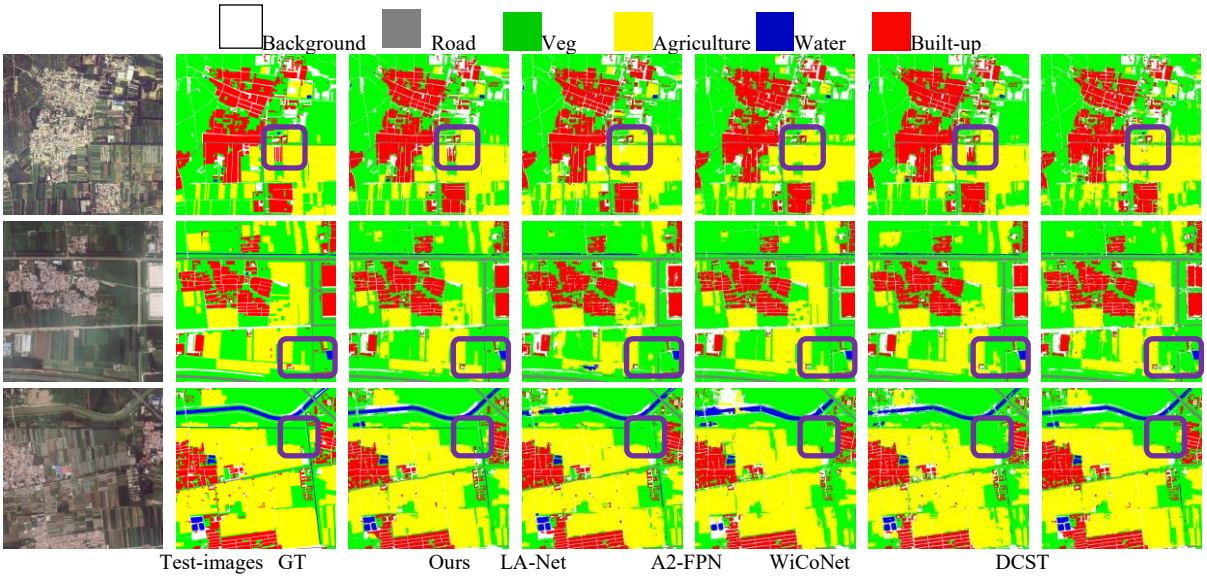


Figure 4. Semantic Segmentation Results Example from the BLU Dataset.

V. Conclusion

In this study, we introduced a novel architecture for the semantic segmentation of remote sensing images, integrating Swin-T and CNNs within a dual encoder setup. Our approach utilizes innovative mechanisms such as the FFM and CAM, which synergistically enhance the discernibility of both local and global features—crucial elements for accurate segmentation. The results from the Beijing Land Use dataset demonstrate that our model outperforms several state-of-the-art methods in terms of segmentation accuracy, as evidenced by improvements in the mIoU. This performance indicates that our method is not only effective but also highly adaptable to the complexities and variability inherent in remote sensing imagery.

In the future, we hope to further explore this architecture across various datasets and environments. There is also an opportunity to enhance the model's efficiency and robustness, achieving higher accuracy and a more detailed understanding of diverse landscape features.

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